

Lab 3: Component Modelling and Architectural Pattern Selection

Hostel Maintenance & Issue Ticketing System

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Architecture Selection and Justification

Architecture Selection: We chose **Layered Architecture** for the Hostel Maintenance & Issue Ticketing System. The system is organized into a Presentation Layer, Business Layer, and Data Layer, with each layer handling a specific responsibility.

Reason 1 – Clear Separation of Concerns: Layered architecture provides a clear separation between the user interface, business operations, and data storage. The Presentation Layer contains the Resident and Maintenance Warden portal. The Business Layer manages ticket creation, staff assignment, work tracking, status updates, SLA monitoring, escalation, and closure verification. The Data Layer stores ticket information and issue photos. This separation makes the system easier to understand, maintain, test, and modify because changes in one layer can be made without unnecessarily affecting the other layers.

Reason 2 – Suitable for Centralized Hostel Management: The system is designed for a centralized hostel-management environment where residents and maintenance staff use a controlled system. Residents can raise and track maintenance tickets, while the Maintenance Warden can review requests, assign maintenance staff, monitor work status, and track deadlines. A layered structure supports this workflow by keeping user interaction separate from the services that process tickets and the repository that stores system data.

Security Advantage: A major security advantage is centralized Authentication and Role-Based Access Control (RBAC). Users receive access according to their roles. Residents can access tickets belonging to their own room, while Wardens and maintenance staff receive permissions appropriate to their responsibilities. Communication can use TLS, and credentials and uploaded issue photos can be stored securely. Centralizing authentication and authorization also makes access control easier to manage consistently across the system.

Performance Benefit: SLA Monitoring and Escalation can operate as background processing within the Business Layer. This allows residents to submit tickets and view their current ticket status without having to wait for SLA checks, escalation logic, or notification processing to finish. The separation of these responsibilities helps keep normal ticket operations responsive while background services handle deadline monitoring and alerts.

Conclusion: Layered Architecture provides a structured design for the Hostel Maintenance & Issue Ticketing System by separating presentation, business logic, and data management. It supports the required ticketing workflow, centralized access control, maintainability, and background SLA processing.